For chief data officers and data platform leaders

Regulators feel the need, the need for speed.

Why the current data management solutions are insufficient for the next wave of regulation.

10s
Euro instant credit transfer window

4h
DORA major-incident initial notification

T+1
EU, UK and Swiss settlement from 11 Oct 2027

2 Dec 2027
AI Act high-risk obligations apply

Executive summary

For two decades, financial institutions have run on batch data. Overnight loads reconciled yesterday's positions, monthly extracts fed regulatory returns and quarterly cycles governed risk reporting. That model is now incompatible with the law. Legislation that has either taken effect since January 2025 or applies by 2028 attaches hard clocks to data: ten seconds to clear a euro payment, four hours to notify a major ICT incident, one business day to settle a trade, and continuous customer screening whenever a sanctions list changes.

The common thread is not speed for its own sake. Each regime requires institutions to make a decision on a single record, at the moment it arrives, against a governed set of rules, and to prove afterwards that the decision was correct. That is a data management problem before it is a compliance problem, and it lands on the Chief Data Officer.

This paper maps the legislation, extracts the data requirements it creates, and sets out the architecture that satisfies them: streaming, incremental ingestion from systems of record into a governed lakehouse, with every record checked against a data contract and a human in the loop. It is the model QBricks was built on.

01 · The default for money

Real-time is already the default for money.

Payments moved first. ACI Worldwide and GlobalData recorded 266.2 billion real-time payment transactions globally in 2023, a 42.2% year-on-year increase, and forecast 575.1 billion by 2028, when more than one in four electronic payments will be real-time. India processed 129.3 billion of those transactions, more than the rest of the top ten markets combined; Brazil's PIX grew 77.9% in a single year; and Oman, Kuwait and Qatar all launched national instant schemes in 2023, making the Middle East the fastest-growing real-time region.

The consequence for data is structural. When value moves in seconds, the controls around it (fraud detection, sanctions screening, liquidity management and reconciliation) must operate on the same clock. A monitoring system that discovers a suspicious pattern in tomorrow's batch has not detected fraud; it has documented it. Regulators have drawn that conclusion and written it into law.

266.2bn
real-time payment transactions globally in 2023

+42.2%
year-on-year growth in real-time payment volume

575.1bn
forecast for 2028, more than one in four electronic payments

02 · The legislative wave

The legislative wave, 2025 to 2028.

Nine instruments across the EU, UK and Asia-Pacific now impose explicit timing or continuity obligations on financial data. The table summarises them; the sections that follow set out what each demands of the data estate.

2.1 Instant Payments Regulation: the ten-second rule

Euro-area banks have been obliged to receive instant euro credit transfers since 9 January 2025 and to send them, with Verification of Payee, since 9 October 2025; payment and e-money institutions follow by 9 April 2027 and non-euro-area PSPs by July 2027. Funds must reach the beneficiary within ten seconds, around the clock.

The regulation's answer to the sanctions problem is instructive. Because ten seconds leaves no time to screen each transaction, Article 5d moves the check to the customer level: PSPs must screen their entire customer base against EU lists at least daily, and immediately whenever a list changes. The obligation is therefore not a faster batch. It is an always-current customer master, continuously reconciled with an external reference dataset, with a record of every alert and action taken.

Nine instruments, nine data clocks

Instrument	Jurisdiction	Key date	Data clock it imposes
Instant Payments Regulation (EU) 2024/886	EU	9 Jan 2025 · 9 Oct 2025	10-second execution, 24/7; Verification of Payee; at-least-daily customer sanctions screening
DORA (EU) 2022/2554 and RTS 2025/301	EU	17 Jan 2025	Initial major-incident notification within 4 hours of classification, 24 hours of awareness
AMLR (EU) 2024/1624	EU	10 Jul 2027	Single rulebook; ongoing monitoring; STR before execution where feasible; harmonised AMLA formats
T+1 settlement (CSDR Art. 5 amendment; UK SI)	EU · UK · CH	11 Oct 2027	Same-day allocation and confirmation; settlement one business day after trade
PSD3 and Payment Services Regulation	EU	Around 2028 (21 months after OJ)	Behavioural transaction monitoring; mandatory fraud-data sharing; VoP on all credit transfers
APP fraud mandatory reimbursement (PSR)	UK	7 Oct 2024	Liability on sending and receiving firm; 5-day reimbursement expectation; inbound monitoring
BCBS 239 and ECB RDARR Guide	Global · EU	In force; ECB priority 2025 to 2027	Risk data aggregated on demand and in stress, with full lineage; Pillar 2 add-ons for laggards
Shared Responsibility Framework (MAS/IMDA)	Singapore	16 Dec 2024 · Jun 2025	Real-time fraud surveillance; real-time alerts; hold rapid account-draining transfers
AI Act (EU) 2024/1689, as amended by Reg. 2026/1744	EU	2 Aug 2026 · 2 Dec 2027	Transparency live; Annex III high-risk (credit scoring, insurance pricing): data governance, automatic logging, human oversight

2.2 DORA: four hours to tell the regulator

Since 17 January 2025, a major ICT-related incident triggers an initial notification within four hours of classification and no later than 24 hours from awareness, an intermediate report at 72 hours and a final report within a month. Classification itself depends on thresholds that cannot be computed from a nightly extract: clients affected above 10% or 100,000, counterparties above 30%, transaction value and data loss. An institution that learns the blast radius of an outage the next morning has already missed the clock.

2.3 AMLR: the single rulebook, July 2027

Regulation (EU) 2024/1624 applies directly in every Member State from 10 July 2027, replacing the transposed directives with one text. AMLA, operational in Frankfurt since July 2025, must publish most of its technical standards by July 2026, and will directly supervise selected cross-border groups from 2028. For the data function the material changes are:

A business-wide risk assessment that is kept current, with documented factors, weightings and board sign-off, including sanctions-evasion risk, rather than an annual document.

Ongoing monitoring against harmonised standards, and suspicious transaction reports filed in AMLA's harmonised format, before the transaction is executed where feasible.

Beneficial ownership at 25% or more, direct or indirect, across every legal entity in the relationship, with the senior managing official as fallback.

Group-wide policies and data sharing, including third-country subsidiaries, and a legal basis in Article 75 for information-sharing partnerships.

Explicit permission for automated individual decision-making, including profiling under GDPR Article 22, provided the safeguards and auditability are in place.

Read together, these clauses describe a system that scores every customer and transaction continuously, against rules that are versioned and auditable, and that can show a supervisor the lineage from raw event to filed report. Five-year retention (extendable to seven) applies to all of it.

2.4 T+1: the post-trade day disappears

The EU, UK and Switzerland move to T+1 securities settlement together on 11 October 2027; the EU legislation was published in the Official Journal on 14 October 2025 and the FCA expects firms to be testing by the end of 2026. In practice all allocations and confirmations must be completed on trade date, and stock loan recalls, collateral and funding must be arranged by early afternoon on T+0. Standing settlement instructions, reference data and positions have to be correct the first time, because there is no second day in which to repair them.

2.5 PSD3 and PSR: fraud liability and mandatory data sharing

Provisionally agreed on 27 November 2025 and endorsed by COREPER on 22 April 2026, the Payment Services Regulation applies roughly 21 months after publication, around 2028. It requires PSPs to operate risk-sensitive, behavioural transaction monitoring, extends Verification of Payee to all credit transfers, shifts liability for impersonation fraud onto the PSP, and creates both the obligation and the legal basis to share fraud data, such as suspicious IBANs and manipulation techniques, between PSPs, telecoms and platforms under GDPR safeguards. Consortium intelligence only works if each participant's contribution is clean, current and governed.

2.6 UK and Singapore: liability regimes that reward detection

The UK's mandatory APP fraud reimbursement regime has been in force since 7 October 2024 and splits liability 50/50 between sending and receiving firms. Frontier Economics' independent review for the PSR, published 1 July 2026, found in-scope losses down about 21% (£73 million a year), nearly 35,000 fewer scams, and firms reimbursing 97% of covered claims, with the largest improvements at the firms that previously had the most fraud. Liability moved the money into detection, and detection means inbound as well as outbound real-time monitoring.

Singapore's Shared Responsibility Framework, effective 16 December 2024, goes further: it names real-time fraud surveillance as a duty. Financial institutions must detect an account being rapidly drained (more than half of a balance of S$50,000 or more within 24 hours) and either block the transaction until the customer confirms or hold it for 24 hours. Breach of the duty means the institution pays. In February 2026 MAS told Parliament that 79% of scam cases now involve authorised transfers, signalling that the surveillance expectation will widen.

£73m
a year lower in-scope APP fraud losses in the UK, about 21%

97%
of covered claims reimbursed by UK firms

79%
of Singapore scam cases now involve authorised transfers

2.7 EU AI Act: the model is only as defensible as its data

The AI Act overlays every regime above, because the monitoring, scoring and screening engines they demand are increasingly AI systems. The Digital Omnibus on AI, Regulation (EU) 2026/1744, in force since 27 July 2026, moved the Annex III high-risk obligations from 2 August 2026 to 2 December 2027. Article 50 transparency duties and the enforcement regime applied from 2 August 2026 as planned, with fines of up to €15 million or 3% of global turnover. Annex III captures creditworthiness assessment and life and health insurance risk pricing directly; AML and fraud models fall under the Act's general transparency and GPAI provisions and are expected to meet the same evidential standard when challenged.

For a high-risk system, Articles 9 to 15 and 26 require of its data what the financial regulations require of transactions:

Data governance (Art. 10). Training, validation and testing datasets must be relevant, representative, free of errors and complete, with documented provenance, collection and preparation, and bias examined and recorded.

Automatic logging (Art. 12). The system must record events throughout its lifetime so that each output can be traced back to the input data and model version that produced it.

Human oversight (Art. 14 and 26). Deployers must assign competent people who can interpret, override or halt the system, and must monitor it in operation, not only at approval.

Continuous risk management (Art. 9). An iterative process across the lifecycle, including post-market monitoring. A model drifting on stale or degraded input data is a compliance event.

The implication is that model governance cannot be separated from data governance. A fraud model with a four-hour decision clock, or a credit model under Annex III, is defensible only if the data feeding it arrived under a contract, was checked on ingestion, and can be replayed record for record. The AI Act's 2 December 2027 date lands within twelve weeks of T+1 and five months after AMLR. The same data estate must satisfy all three.

Three deadlines, one data estate

10 Jul 2027	AMLR applies in every Member State: ongoing monitoring, harmonised STR formats, five-to-seven-year retention.
11 Oct 2027	T+1 settlement in the EU, UK and Switzerland: allocations and confirmations on trade date.
2 Dec 2027	**AI Act Annex III high-risk obligations: data governance, automatic logging, human oversight.**

03 · The requirement

What the legislation actually requires of data.

Strip away the sector-specific language and the instruments converge on five capabilities. Each is a property of the data platform, not of a compliance application bolted on top of it.

01 · Latency

Decide on arrival

Seconds (IPR, SRF), hours (DORA) or same day (T+1). Data must be usable the moment it is created, not after the next load.

02 · Incremental

Every change, once

Sanctions deltas, customer updates and transactions are streams of change. Full reloads are too slow and destroy the audit trail.

03 · Governed

Rules before records

AMLR, BCBS 239, the ECB Guide and AI Act Art. 10 expect documented definitions, provenance and quality thresholds applied at ingestion, not reconciled later.

04 · Auditable

Lineage end to end

Five-to-seven-year retention, before-and-after state, and a traceable path from source event to STR or risk report.

05 · Accountable

Human in the loop

Automated decisions are permitted (AMLR says so explicitly, and the AI Act mandates oversight) only with safeguards, override and documented review.

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global systemically important banks fully compliant with BCBS 239 in the Basel Committee's 2023 progress report.

The industry's track record against the oldest of these expectations shows how far there is to go. More than a decade after BCBS 239 was published, the Basel Committee's 2023 progress report found only 2 of 31 global systemically important banks fully compliant, with no principle fully implemented across all banks. The ECB has responded with its May 2024 RDARR Guide, made remediation a supervisory priority for 2025 to 2027 and signalled Pillar 2 capital add-ons and fines for institutions that continue to rely on manual workarounds and cannot give credible target dates. A February 2025 ECB newsletter named the recurring failure modes: no recent gap analysis, weakly controlled manual processes, and incomplete lineage.

Those failures are architectural. A programme that documents lineage from the aggregation layer to the report, while the pipelines feeding the aggregation layer remain thousands of hand-written notebooks, has not solved the problem that the supervisor is asking about.

04 · The architecture

The architecture that satisfies the clock.

QBricks was designed from the requirements above rather than retrofitted to them. It is a streaming data management platform that moves data from systems of record into a governed lakehouse incrementally, change by change, and digitally compares every record against a pre-agreed governance framework before it lands. Four design choices map directly onto the regulatory drivers.

Streaming, incremental ingestion

Each change is captured and delivered as it occurs, so sanctions screening, fraud monitoring and DORA impact assessment run on current state.

Answers the IPR, SRF and DORA latency requirements.

Data contracts (ODCS)

Definitions, ownership, quality rules and retention are expressed in the Open Data Contract Standard and enforced at ingestion, in Unity Catalog or the equivalent on Fabric and Snowflake.

Answers AMLR's documented risk methodology and BCBS 239 Principles 2 to 6.

Before-and-after files, full lineage

Every accepted and rejected record is retained with its prior state and the rule that decided it. A supervisor's request, an STR, a T+1 settlement break or a DORA final report can be traced to source without reconstruction.

Agentic metadata, human in the loop

Agents learn and improve metadata and quality rules over time while remaining isolated from the data lake; a person approves changes.

The safeguarded automation AMLR and FinCEN's 2026 proposal encourage, and the human oversight AI Act Articles 14 and 26 require.

The economics follow from the architecture. Replacing hand-built pipelines with contract-driven ingestion means that each new regulatory obligation, whether a new AMLA reporting format, a new VoP match rule or a new SRF threshold, is a change to a contract, not a new engineering project. From system of record to lakehouse in hours, not quarters.

Priorities for the next 18 months.

01 **Inventory the clocks.** For each obligation in Section 2, record the data domains it touches, the latency it demands and the current end-to-end latency of that data. The gap is the programme.

02 **Move customer and reference masters to continuous update first.** Sanctions screening under the IPR, beneficial ownership under AMLR and settlement instructions under T+1 all fail on stale masters before they fail on transaction speed.

03 **Put contracts in front of pipelines.** Define ODCS contracts for the regulatory-critical domains and enforce them at ingestion. This is the single step that converts BCBS 239 remediation from a documentation exercise into a control.

04 **Retain before-and-after state by default.** Lineage that can be replayed is the only lineage a supervisor will accept in an on-site inspection.

05 **Classify your models now, and trace their data.** Inventory every AI system against Annex III, and for each high-risk system establish dataset provenance and automatic logging before 2 December 2027. The governance in Article 10 is a property of the ingestion layer, not the model.

06 **Test to the 2026 milestones.** AMLA's technical standards (July 2026), FCA T+1 readiness (end 2026) and PSR publication (mid 2026) are the checkpoints by which the architecture must be in place, not merely planned.

The institutions that will clear the 2027 deadlines comfortably are not those with the largest compliance budgets. They are the ones whose data arrives governed, in real time, from the system of record, so that each new regulation is a rule to add, not a platform to build.

No more data pipelines.

See how QBricks takes a regulated data domain from system of record to a governed lakehouse in hours, with every record contract-checked and a human in the loop.

Request a briefing

Sources

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Dates and figures are as published by the cited sources at the time of writing (October 2026). Instruments still in the legislative process (PSD3 and PSR, FinCEN programme rule) may change before application. This paper is not legal advice. Companion paper: Why querying enterprise data at scale should not cost a fortune (QBricks, October 2026).

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